

Hands-on Lab: Create Tables and Load Data in MySQL using phpMyAdmin



Estimated time needed: 20 minutes
In this lab, you will learn how to create tables and load data in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

Objectives

- After completing this lab, you will be able to use phpMyAdmin with MySQL to:
- Create a database.
 - Create tables.
 - Load data into tables manually using the phpMyAdmin GUI.
 - Load data into tables using a text/script file.

Software Used in this Lab

In this lab, you will use [MySQL](#). MySQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.



To complete this lab you will utilize MySQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Database Used in this Lab

Books database has been used in this lab.

The following diagram shows the structure of the **myauthors** table from the Books database:

myauthors	
author_id	int
first_name	varchar(200)
middle_name	varchar(50)
last_name	varchar(100)

In the table, **author_id** is an integer, **first_name** is a string that stores a maximum of 100 characters, **middle_name** is a string that stores a maximum of 50 characters, and **last_name** is a string that stores a maximum of 100 characters.

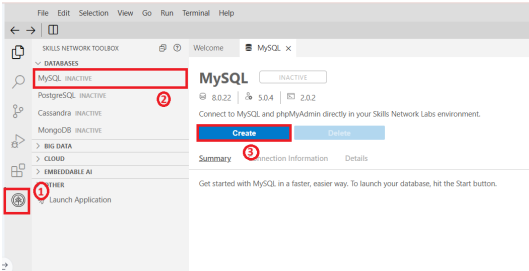
Task A: Create a database

Start the MySQL service session using the `start mysql` in IDE `!start` directive.

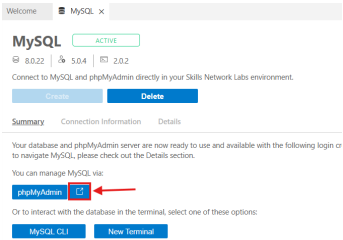
[Open MySQL Page in IDE](#)

If the icon doesn't start the MySQL database, follow the steps below.

1. Click the Skills Network extension button on the left side of the window.
2. Open the DATABASES menu and click MySQL.
3. Click Create. MySQL may take a few moments to start.



4. Open the phpMyAdmin tool in a new tab in your browser:



5. You will see the phpMyAdmin GUI tool.

phpMyAdmin

Recent

Favorites

New

information_schema

mysql

performance_schema

sakila

sys

Server: mysql:3306

DatabasesSQLStatusUser accountsExportImportSettingsBinary logReplicationVariablesCharsetsEnginesP

General settings

Server connection collation: utf8mb4_unicode_ci

More settings

Appearance settings

Language: English

Theme: pmahomme

Database server

Server: mysql via TCP/IP

Server type: MySQL

Server connection: SSL is not being used

Server version: 8.0.22 - MySQL Community Server - GPL

Protocol version: 10

User: root@172.18.0.2

Server charset: UTF-8 Unicode (utf8mb4)

Web server

Apache/2.4.38 (Debian)

Database client version: libmysql - mysqlnd 7.4.15

PHP extension: mysqli curl mbstring

PHP version: 7.4.15

phpMyAdmin

Version information: 5.0.4, latest stable version: 5.1.0

Documentation

Official Homepage

Contribute

Get support

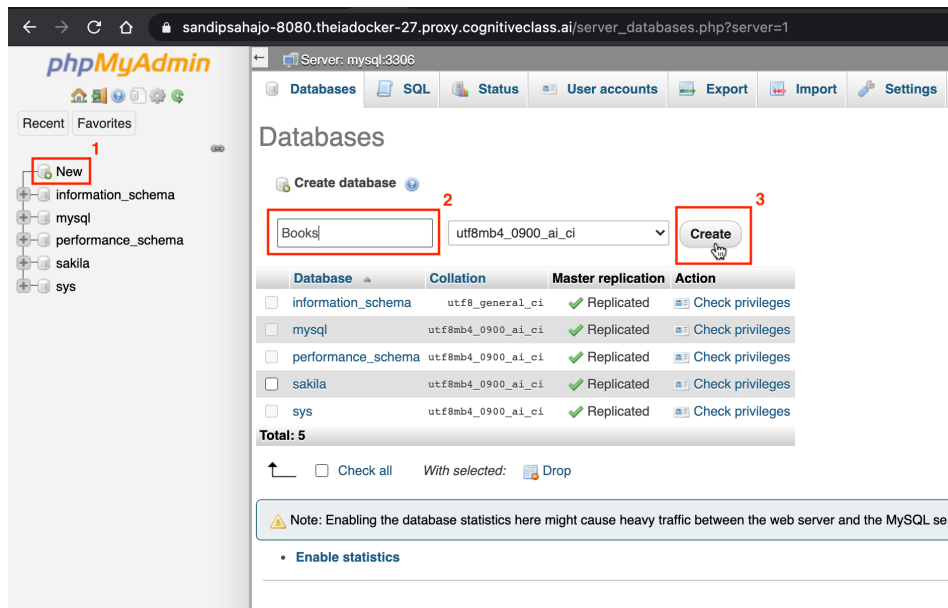
List of changes

License

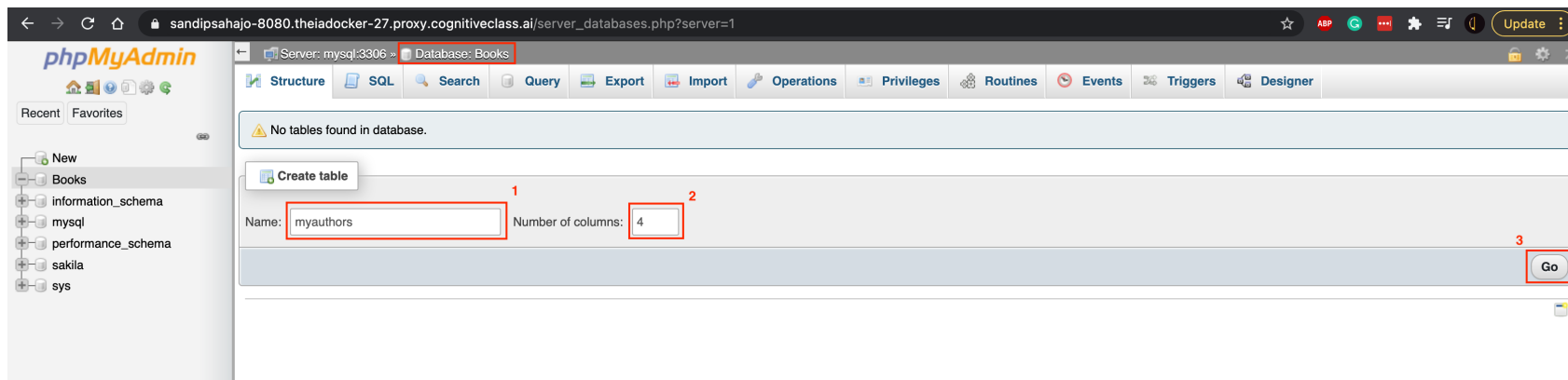
6. In the tree-view, click **New** to create a new empty database. Then enter **books** as the name of the database and click **Create**.
The encoding will be left as **utf8mb4_unicode_ci**. UTF-8 is the most commonly used character encoding for content or data.

2 of 7

1/1/2026, 3:54 PM

**Task B: Create tables**

1. In the Create table interface for the empty database **Books**, enter **myauthors** as the table name and **4** for the Number of columns. This is the first step to creating the table **myauthors** that was shown earlier in this lab. Then click **Go**.



2. Enter the table definition for the **myauthors** table as shown in the image below with highlighted boxes. Then click **Save**.

砂砂砂砂砂-8080.theiadocker-27.proxy.cognitiveclass.ai/sql.php?db=Books&table=myauthors

Server: mysql:3306 » Database: Books » Table: myauthors

Table name: myauthors Add 1 column(s) Go

Name	Type	Length/Values	Default	Collation	Attributes	Null	Index	A.I	Comments	Virtuality
author_id	INT		None			☐	---	☐		
first_name	VARCHAR	100	None			☐	---	☐		
middle_name	VARCHAR	50	None			☐	---	☐		
last_name	VARCHAR	100	None			☐	---	☐		

Structure

Table comments: Collation: Storage Engine: InnoDB

PARTITION definition:

Partition by: (Expression or column list)

Partitions:

Preview SQL Save

3. The Table structure for the myauthors table will appear.

砂砂砂砂砂-8080.theiadocker-27.proxy.cognitiveclass.ai/tbl_structure.php?db=Books&table=myauthors

Server: mysql:3306 » Database: Books » Table: myauthors

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	author_id	int			No	None			Change Drop More
☐ 2	first_name	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐ 3	middle_name	varchar(50)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐ 4	last_name	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More

Check all With selected: Browse Change Drop Primary Unique Index Fulltext

Print Move columns Normalize

Add 1 column(s) after last_name Go

Task C: Load data into tables manually using the phpMyAdmin GUI

1. Sometimes, you may want to load a few data rows of data, but you may not have a SQL script on hand to do that. In this case, you can manually load the data into phpMyAdmin. Since this is a manual process, it is better for inserting a small amount of data rather than a large amount.

To load data manually, go to the **Insert** tab for the **myauthors** table. Enter data for 2 rows of the **myauthors** table as shown in the image below with highlighted boxes. Then click **Go** at the bottom.

phpMyAdmin

Recent Favorites

Server: mysql:3306 - Database: Books - Table: myauthors

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Column	Type	Function	Null	Value
author_id	int			1
first_name	varchar(100)			Merritt
middle_name	varchar(50)			
last_name	varchar(100)			Eric

Go

☐ Ignore

Column	Type	Function	Null	Value
author_id	int			2
first_name	varchar(100)			Linda
middle_name	varchar(50)			
last_name	varchar(100)			Mui

Go

Insert as new row and then Go back to previous page

Preview SQL Reset Go

2. Notification of the successful insertion of 2 rows to the **myauthors** table will appear.

✓ 2 rows inserted.

```
INSERT INTO `myauthors` (`author_id`, `first_name`, `middle_name`, `last_name`) VALUES ('1', 'Merritt', '', 'Eric'), ('2', 'Linda', '', 'Mui');
```

3. Go to the **Browse** tab for the **myauthors** table to check the newly inserted rows.

phpMyAdmin

Server: mysql:3306 - Database: Books - Table: myauthors

Browse Structure SQL Search Insert Export Import

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are disabled.

Showing rows 0 - 1 (2 total, Query took 0.0004 seconds.)

`SELECT * FROM `myauthors``

☐ Show all | Number of rows: 25 | Filter rows: Search this table

+ Options

author_id	first_name	middle_name	last_name
1	Merritt		Eric
2	Linda		Mui

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Task D: Load data into tables using a text/script file

1. Now you will use a SQL script to import the remainder of the **myauthors** table data. A SQL script file contains commands and statements that perform operations on your database, and can be useful when importing a large amount of data.

Download the SQL script below to your local computer:

[mysql_table-myauthors_insert-data.sql](#)

2. Go to **Import** tab for the **myauthors** table. Click **Choose File** and load the **mysql_table-myauthors_insert-data.sql** file from your local computer storage. The rest of the settings can be left as they are because you are importing a SQL script that is encoded with UTF-8.

Then click **Go**. Notification of import success will appear.

phpMyAdmin

Server: mysql:3306 - Database: Books - Table: myauthors

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Importing into the table "myauthors"

File to import:

File may be compressed (gzip, bzip2, zip) or uncompressed.
A compressed file's name must end in `.[format][compression]`. Example: `.sql.zip`

Browse your computer: **Choose File** mysql_table...sert-data.sql (Max: 2,048KIB)

You may also drag and drop a file on any page.

Character set of the file: utf-8

Partial import:

☒ Allow the interruption of an import in case the script detects it is close to the PHP timeout limit. *(This might be a good way to import large files, however it can break transactions.)*

Skip this number of queries (for SQL) starting from the first one: 0

Other options:

☒ Enable foreign key checks

Format:

SQL

Format-specific options:

SQL compatibility mode: NONE

☒ Do not use `AUTO_INCREMENT` for zero values

Go

Import has been successfully finished, 1376 queries executed. (mysql_table-myauthors_insert-data.sql)

3. Go to the **Browse** tab for the **myauthors** table again to check the newly inserted rows appear along with previously inserted 2 rows.

phpMyAdmin

Recent Favorites

New Books New myauthors information_schema mysql performance_schema sakila sys

Browse Structure SQL Search Insert Export Import

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete feature

Showing rows 0 - 24 (1378 total, Query took 0.0003 seconds.)

SELECT * FROM `myauthors`

1 > >> | Number of rows: 25 Filter rows: Search this table

+ Options

author_id	first_name	middle_name	last_name
1	Merritt		Eric
2	Linda		Mui
3	Alecos		Papadatos
4	Paul	C.van	Oorschot
5	David		Cronin
6	Richard		Blum
7	Yuval	Noah	Harari
8	Paul		Albitz
9	David		Beazley
10	John	Paul	Shen
11	Andrew		Miller
12	Melanie		Swan
13	Neal		Ford
14	Nir		Shavit
15	Tim		Kindberg
16	Mike		McQuaid
17	Brian	P.	Hogan
18	Jean-Philippe		Aumasson
19	Lance		Fortnow
20	Richard	C.	Jeffrey
21	William	L.	Simon
22	Magnus	Lie	Hetland
23	Mike		McShaffry
24	Norman		Matioff
25	John	E.	Hopcroft

1 > >> | Number of rows: 25 Filter rows: Search this table

Congratulations! You have completed this lab, and you are ready for the next topic.

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Other Contributor(s)

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